

Gold Price Analysis (1970–2026)

Excel Observations

1. Data Cleaning

The original dataset contained 20,819 rows. 307 completely blank rows were removed. No duplicate records were identified. The final cleaned dataset contains 20,512 valid records with no missing values. The analysis period covers 01 January 1970 to 27 February 2026.

2. Overall Gold Price

The average closing gold price is approximately \$5,576.05. The minimum closing price was \$31.54 on 22 February 1970. The maximum closing price was \$48,805.41 on 09 June 2024. The large difference between the minimum and maximum values indicates substantial long-term growth in gold prices.

3. Yearly Performance

Gold prices remained comparatively low during the earlier decades of the dataset. A stronger upward movement is visible from approximately 2005 onwards. 2024 recorded the highest annual average closing price, at approximately \$43,058.87. The next highest annual average prices were observed in 2025 (\$41,890.87), 2026 (\$37,575.75, partial year), and 2023 (\$34,820.09).

4. Monthly Performance

The monthly averages are relatively close compared with the much larger differences observed across years. The monthly analysis helps identify seasonal differences in average closing prices.

5. Daily Price Movement

The dataset contains both increasing and decreasing daily closing prices. Based on the calculated close-to-close movement, there are approximately 10,532 increase days and 9,979 decrease days, with one day showing no change. This indicates that daily price movements occur in both directions rather than following a consistently increasing pattern.

6. Price Volatility

The difference between annual high and low prices becomes much larger in recent years. 2024 had the largest annual high-low range, approximately \$12,059.08. This indicates considerably larger absolute price fluctuations in recent years compared with earlier periods.

7. Trading Volume

The average daily trading volume is approximately 299,144. Trading volume varies considerably across the dataset and generally becomes larger in the more recent period. Yearly volume analysis can be used to identify periods of increased market activity.

8. USD Index Relationship

Gold closing price and the USD Index show a strong negative correlation of approximately -0.975 in this dataset. This means that the two variables tend to move in opposite directions over the analyzed observations. Correlation indicates an association and should not be interpreted as proof that changes in the USD Index directly cause changes in gold prices.

9. Recent Gold Price Trend

The most significant increase in average gold prices occurs in the recent years of the dataset. The annual average increased from approximately \$12,880 in 2017 to \$43,059 in 2024, showing a substantial increase over this period.

10. Overall Observation

The Excel analysis shows a strong long-term upward trend in gold prices, increasing price fluctuations in recent years, changing trading activity, and a strong inverse statistical relationship between gold prices and the USD Index.

Short Observation for Project Report

After data cleaning, 20,512 valid records remained from the original 20,819 rows, with 307 blank rows removed and no duplicate or missing records. The average closing gold price is \$5,576.05, while the minimum and maximum closing prices are \$31.54 and \$48,805.41 respectively. Gold prices show a strong long-term upward trend, particularly after 2005, with 2024 recording the highest annual average closing price of approximately \$43,058.87. Recent years also show greater price fluctuations and trading activity. The analysis indicates a strong negative correlation of approximately -0.975 between gold price and the USD Index.

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Power BI Observations

Power BI Project Structure

Page 1: Gold Price Overview

- KPI Cards
- Daily Close Price line chart
- Average Gold Price by Year
- Average Gold Price by Month
- Year / Quarter / Month slicers

Page 2: Trend and Volatility

- Open vs Close
- Annual Price Range
- Annual Change %
- Volatility
- Top increase/decrease years

Page 3: Market Relationship

- Gold Close vs USD Index scatter
- Gold Price vs USD Index trend
- Yearly Trading Volume
- Maximum/Minimum price cards

Step 1: Open Power BI

Open **Power BI Desktop**.

Select:

Home → Get Data → Excel workbook

Choose:

Gold_Price_Mini_Project_Dashboard_Pivot.xlsx

You should see sheets such as:

- Cleaned Data
- Analysis Formulas
- Pivot Tables
- Dashboard

For the actual Power BI report, select:

Cleaned Data

Then click:

Load

Step 2: Check your columns

Your Cleaned Data table contains the important fields:

Column	Use
Date	X-axis / Date filter
Open	Open price
High	High price
Low	Low price
Close	Main gold price
Volume	Trading volume
USD_Index	USD relationship
Year	Year slicer
Month	Month analysis
Month Name	Month chart
Quarter	Quarter slicer
Day	Day analysis
Day of Week	Day analysis
Year-Month	Monthly trend
Price Change	Price movement

Column	Use
Change %	Percentage movement
Price Range	High - Low
Close Change (Formula)	Daily change
Daily Return % (Formula)	Daily return
Price Direction (Formula)	Increase/Decrease
Significant Move (Formula)	Significant/Normal

Step 3: Create Page 1

Rename the first page:

Gold Price Overview

At the bottom of Power BI, double-click:

Page 1

Rename it.

3.1 Add the title

Go to:

Insert → Text box

Type:

GOLD PRICE ANALYSIS (1970–2026)

Below it:

Page 1: Gold Price Overview

Format the title with a dark-blue background and gold/white text.

Step 4: Create KPI Cards

We need approximately **5 KPI cards** on Page 1.

Card 1: Total Records

Insert:

Card

Drag:

Date

into the card.

Change summarization to:

Count

This should show approximately:

20,512

Title:

Total Records

Card 2: Average Gold Price

Insert another Card.

Drag:

Close

Set:

Average

Result:

5,576.05

Title:

Average Close Price

Card 3: Maximum Gold Price

Insert Card.

Drag:

Close

Set:

Maximum

Result:

48,805.41

Title:

Highest Price

Card 4: Minimum Gold Price

Insert Card.

Drag:

Close

Set:

Minimum

Result:

31.54

Title:

Lowest Price

Card 5: Average Daily Return

Insert Card.

Drag:

Daily Return %

Set:

Average

Format it as:

Percentage

Title:

Average Daily Return

Step 5: Create the Year / Quarter / Month slicers

This is important because your assignment specifically asks for slicers.

Go to:

Insert → Slicer

Slicer 1

Field:

Year

Slicer 2

Field:

Quarter

Slicer 3

Field:

Month Name

Arrange them on the left or top.

Your dashboard will look like:

Year [All ▼]

Quarter [All ▼]

Month [All ▼]

Step 6: Daily Close Price Chart

This is the main chart on Page 1.

Select:

Line chart

X-axis

Date

Y-axis

Close

Set aggregation:

Average or **Don't summarize**, depending on the visual behavior.

For daily data, use:

Date → Date

not Date Hierarchy if you want the complete daily timeline.

Chart title:

Daily Gold Closing Price Trend (1970–2026)

You should see the long-term gold price movement from 1970 to 2026.

Step 7: Average Gold Price by Year

Add a:

Clustered column chart

X-axis

Year

Y-axis

Close

Aggregation:

Average

Title:

Average Gold Price by Year

This will show the strong increase in average gold price in recent years.

Step 8: Average Gold Price by Month

Add:

Clustered column chart

X-axis

Month Name

Y-axis

Close

Aggregation:

Average

Important:

Sort the month names chronologically.

Select Month Name in the Data view and use:

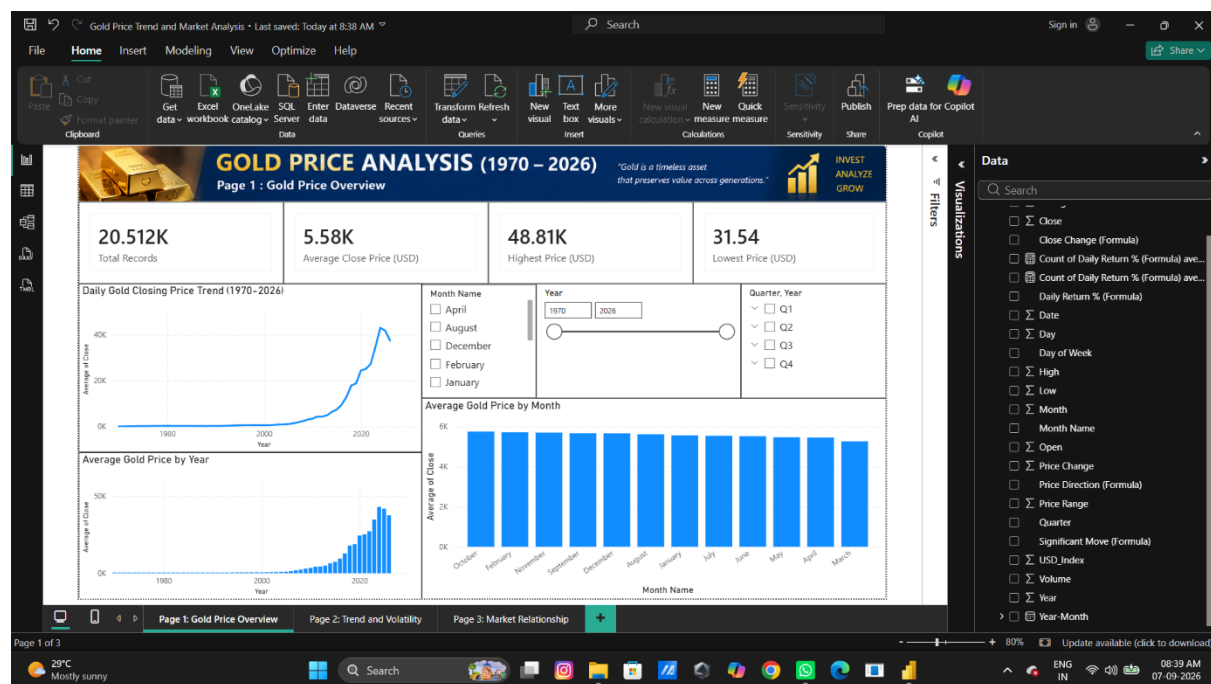
Column tools → Sort by column → Month

This prevents Power BI from sorting months alphabetically.

Title:

Average Gold Price by Month

Page 1 layout



Page 2: Trend and Volatility

Create a new page using:

+

Rename:

Trend and Volatility

Chart 1: Open vs Close

Select:

Line chart

X-axis

Date

Y-axis

Open

and

Close

You will get two lines.

Title:

Open vs Close Price Trend (1970–2026)

Chart 2: Annual Price Range

Your dataset already has:

Price Range

which represents:

High - Low

Create a:

Column chart

X-axis

Year

Y-axis

Price Range

Aggregation:

Average

Title:

Annual Price Range (High - Low)

Chart 3: Change %

Create:

Column chart

X-axis

Year

Y-axis

Change %

Aggregation:

Average

Turn on:

Data colors

You can use:

- Positive = green
- Negative = red

Title:

Yearly Change % in Close Price

Chart 4: Volatility

For a simple Power BI version, use:

Price Range

by Year.

Alternatively, create a DAX measure for annual standard deviation.

Go to:

Modeling → New measure

Use:

Annual Price Volatility =

STDEV.P('Cleaned Data'[Close])

Then create:

Column chart

X-axis:

Year

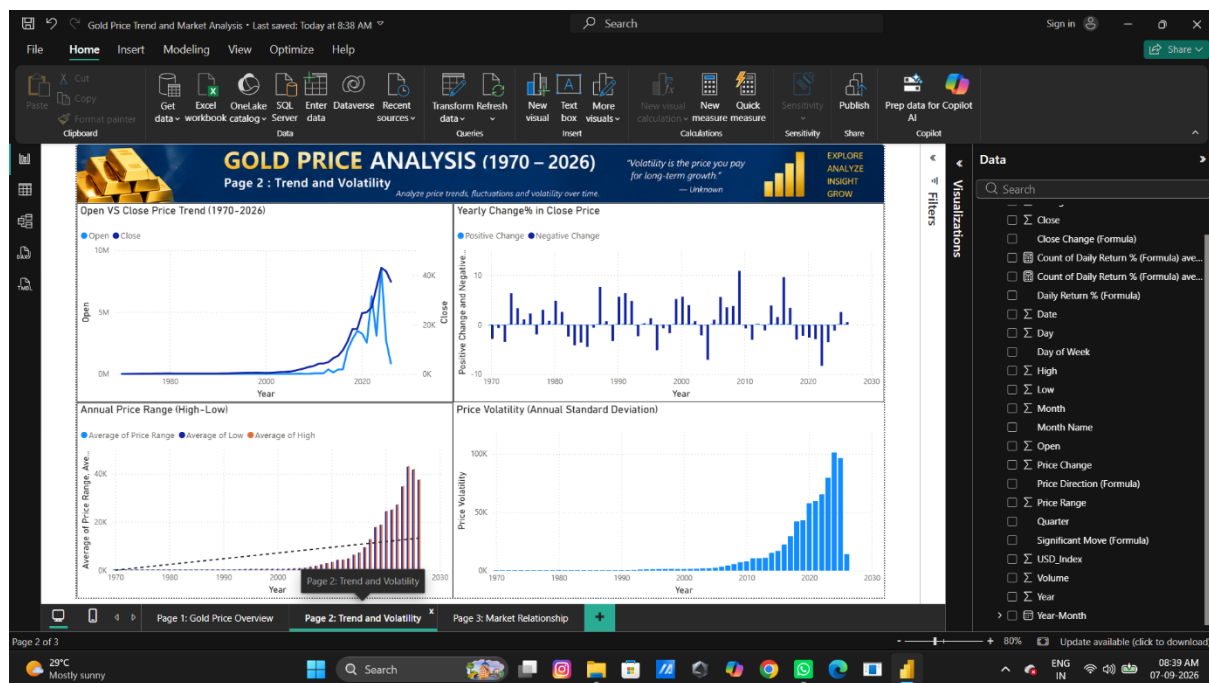
Y-axis:

Annual Price Volatility

Title:

Price Volatility by Year

Page 2 layout



Page 3: Market Relationship

Create another page:

+ → Rename → Market Relationship

This page is particularly important because your dataset contains USD_Index.

Step 9: Maximum Price Card

Create a Card using:

Close

Aggregation:

Maximum

You can display:

48,805.41

But you also want the date.

For that, create a DAX measure.

Go to:

Modeling → New measure

Maximum Price =

MAX('Cleaned Data'[Close])

Then:

Maximum Price Date =

VAR MaxPrice =

MAX('Cleaned Data'[Close])

RETURN

CALCULATE(

MAX('Cleaned Data'[Date]),

'Cleaned Data'[Close] = MaxPrice

)

This gives the date associated with the maximum price.

Step 10: Minimum Price

Create:

Minimum Price =

MIN('Cleaned Data'[Close])

And:

Minimum Price Date =

VAR MinPrice =

MIN('Cleaned Data'[Close])

RETURN

CALCULATE(

MIN('Cleaned Data'[Date]),

'Cleaned Data'[Close] = MinPrice

)

Your KPI section can show:

Maximum Price

\$48,805.41

09 Jun 2024

and

Minimum Price

\$31.54

06 Dec 1971

Step 11: Gold Price vs USD Index

This is the most important visual on Page 3.

Select:

Scatter chart

Use:

X-axis

USD_Index

Y-axis

Close

Title:

Gold Closing Price vs USD Index

You should see a strong downward relationship.

Your dataset shows approximately:

Correlation = -0.975

That means gold price and the USD Index have a very strong negative linear relationship in this dataset.

Step 12: Create the correlation measure

Power BI doesn't have a simple built-in correlation measure, so create:

Modeling → New measure

Gold USD Correlation =

VAR MeanGold =

AVERAGE('Cleaned Data'[Close])

VAR MeanUSD =

AVERAGE('Cleaned Data'[USD_Index])

VAR Numerator =

SUMX(
 'Cleaned Data',
 ('Cleaned Data'[Close] - MeanGold) *
 ('Cleaned Data'[USD_Index] - MeanUSD)
)

VAR Denominator =

SQRT(
 SUMX(
 'Cleaned Data',
 ('Cleaned Data'[Close] - MeanGold)^2
)
 *
 SUMX(
 'Cleaned Data',
 ('Cleaned Data'[USD_Index] - MeanUSD)^2
)
)

RETURN

DIVIDE(Numerator, Denominator)

Add this measure to a Card.

It should show approximately:

-0.975

Format it to 3 decimal places.

Step 13: Gold Price and USD Index Trend

Create a:

Line chart

X-axis:

Year

For the values, you need:

Close

and

USD_Index

Because these have very different scales, use a **Line and clustered column chart** or a visual with a secondary Y-axis.

Title:

Gold Price and USD Index Trend (1970–2026)

Step 14: Yearly Trading Volume

Create:

Column chart

X-axis

Year

Y-axis

Volume

Aggregation:

Sum

Title:

Yearly Trading Volume (1970–2026)

This will show the increase in trading volume over time.

